

BASS / HTT Departure-Decomposition Programme

Research-Progress Summary and Real-Data Analysis Plan
(Planck raw-data E2E + the pre-solver joint PV+CMB analysis)

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Abstract

This report summarises the current state of the BASS/HTT FLRW-departure programme and sets out the plan for the real-data analyses that can be run *before* a native low- ℓ anisotropic Boltzmann solver (“BASS”) exists. The programme delivers a *diagnostic, provenance-bound* comparator over four anisotropy sectors $g = (\Sigma^2, W^2, \Omega_{\text{tilt}}, \Omega_k)$, a suite of conditional EGS-type (Ehlers–Geren–Sachs) theorems, and a growing set of model-independent real-data descriptors. Since the previous revision it has been upgraded on two fronts that close a genuine vulnerability of the earlier plan. (i) *Axis C*: a local observer boost leaks an $O(\beta^2)$ kinematic quadrupole into the shear sector and could fake a $\Sigma^2 > 0$; a closed-form **kinematic deprojection** $\tilde{\Sigma}^2 = \Sigma^2 - \alpha(\Omega_{\text{tilt}})^2$ now makes the low- ℓ shear reading provably immune to it (gate- and Wolfram-verified). (ii) *Axis D*: the proposed joint likelihood $\ln \mathcal{L}_{\text{Total}} = \ln \mathcal{L}_{\text{PV}} + \ln \mathcal{L}_{\text{CMB}}$ is critically evaluated and ported to what is honestly runnable now — a **feasible correlated-covariance bulk-flow (tilt) analysis** on real CF4 (low-rank Woodbury, not a dense $38k \times 38k$ inversion), a **real off-diagonal BipoSH statistical-isotropy measurement** on the Planck SMICA map, and a **JWST distance-anchor acquisition pipeline** whose prior, through the Axis-C coupled Fisher, breaks the Σ^2 – Ω_{tilt} degeneracy. The CMB *theory* sector — the mode-coupled $C_{\ell m, \ell' m'}(g)$ prediction for a Bianchi g — is the native solver and is kept **fail-closed**. The programme does *not* establish a Bianchi-family identification, an anisotropic geometry, a native-solver validation, a physical-vorticity detection, or a globally significant low- ℓ anomaly. The Planck-raw E2E analysis of §7 remains the matched null that would upgrade the K1 morphology statistic; its full pipeline is implemented and validated, awaiting only the manual ~ 1 TB download.

1 Programme and framework

The programme tests whether the cosmic microwave background (CMB) and low-redshift peculiar velocities depart from the standard isotropic FLRW cosmology toward an anisotropic (Bianchi + tilt) one. It deliberately works *without* a native low- ℓ anisotropic Boltzmann solver (which does not yet exist in-tree), so it makes no family-identification or geometry-detection claim. The primary object is the **graded comparator vector**

$$g = (\Sigma^2, W^2, \Omega_{\text{tilt}}, \Omega_k) \in \mathbb{R}^4, \quad x_C = \langle c, g \rangle, \quad c = (+1, -1, +1, +1), \quad (1)$$

where Σ^2 is the shear, W^2 the vorticity, Ω_{tilt} the tilt, and Ω_k the anisotropic-curvature sector; the signed scalar x_C is a *derived* linear summary, so the per-sector content is preserved and the sign-cancellation pathology of a single number is avoided. An equivalent representation casts the comparator as a positive-semidefinite matrix $M = \text{diag}(g) \succeq 0$ with $x_C = \text{tr}(CM)$ (bit-identical), whose admissible set is a convex PSD cone and whose reachable eigen-directions encode identifiability. Two velocity-like sectors — the observer’s local boost β (an $\ell = 1$ frame effect, *not* a sector of g) and the global tilt Ω_{tilt} (a genuine bulk flow) — overlap on the sky; keeping them apart is the subject of Axis C (§3).

Honest claim envelope. All results are diagnostic-only / model-independent / conditional-theorem content. No data-fitting headline exceeds the registry-and-shear-source-validated envelope; the eight Bianchi families beyond {FLRW, I, V, IX} are restricted to axis-aligned mode subsets, and any family/geometry statement remains gated behind the (unbuilt) native solver and its full gate bundle. Σ^2 on the real sky stays **partial**.

2 Theorem programme (progress)

The conditional EGS-type theorem suite spans three axes — mathematics/ statistics, GR/covariant-Boltzmann, and data interpretation — and is gate- and (where stated) Wolfram-verified. The consolidated ledger (`scripts/build_egs_results_table.py`) now holds **25 rows: 19 proven now** (6 closed-form symbolic, 13 gate-verified) and 6 real-data rows, summarised in Table 1. `make_egs3-gates` runs 61 tests. The rows added since the previous revision are the kinematic-deprojection cluster (EGS3-C1–C4, Axis C) and the BASS-Extended joint PV+CMB cluster (EGS3-D1–D4, Axis D).

Table 1: Consolidated theorem/result ledger (status counts: 6 `proven_symbolic`, 13 `proven_gate`, 1 `measured_synth`, 1 `measured`, 2 `measured_partial`, 1 `measured_no_go`, 1 `blocked`).

Item	Axis	Statement (abbreviated)
NT-A1	Math/Stat	Closure-conditional quadrupole-filling identity ($\kappa = 4/21$)
NT-A3	Math/Stat	Single-sky sampling dispersion $\sqrt{2/5}$ (one estimator)
EGS3-A1	Math/Stat	Graded rank-2 identifiability; null $\{W^2\}$ (structural)+ $\{\Omega_k\}$ (no-channel)
EGS3-A3	Math/Stat	Π as a calibrated e-value (Markov bound; GRF- Λ CDM null)
EGS3-A4	Math/Stat	Rao-Blackwell sufficiency of the reachable-sector statistic
NT2-A1	Math/Stat	Multi- ℓ Fisher floor conditional on the registered shear response
NT2-A2	Math/Stat	Octupole information saturation (tail beyond $\ell \sim 20$ converges)
EGS3-B1	GR/Boltz	Semi-native transfer: floor is a single-mode, finite- k profile
EGS3-B2	GR/Boltz	Volterra depth-memory of the tilt stress; $\equiv$ ODE + Grönwall
NT2-B1	GR/Boltz	Two-sided shear bracket within the registered closure/H3 model
NT2-B3/EGS3-B3	GR/Boltz	Radial-velocity vorticity blindness; transverse channel re-opens
EGS3-B4	GR/Boltz	Covariant bracket constants ($C_{\text{up}}\kappa(1+R) > 1$)
EGS3-PSD	Math/Stat	PSD-cone representation, $x_C = \text{tr}(CM)$, bit-identical
EGS3-C1	Math/Stat	Kinematic deprojection $\tilde{\Sigma}^2 = \Sigma^2 - \alpha(\Omega_{\text{tilt}})^2$, boost-immune
EGS3-C2	Math/Stat	Coupled Fisher $F_{\Sigma^2, \Omega_{\text{tilt}}} \propto \beta^2$; inflation $1/(1-r^2)$
EGS3-C3	Math/Stat	Boost/tilt separability: rank 2 generic, degenerate iff axis-aligned
EGS3-C4	Math/Stat	Injection-recovery: naive Σ^2 FPR high, deprojected FPR nominal
EGS3-D1	Data	CF4 Woodbury tilt $ \mathbf{B} \simeq 341$; JWST Ω_{tilt} -prior forecast (<code>measured_synth</code>)
EGS3-D2	Math/Stat	PV prior reduces Σ^2 inflation — degeneracy break (<code>proven_gate</code>)
EGS3-D3	Data	Real SMICA/Commander BipoSH SI, $L=1, 2$, $p \simeq 0.68$ (<code>measured_partial</code>)
EGS3-D4	Data	Theory- g $C_{\ell m, \ell' m'}(g)$ fail-closed (<code>blocked</code>)
K1	Data	Low- ℓ morphology look-elsewhere p (<code>measured_partial</code>)
K5	Data	CF4 bulk-flow amplitude + conditional coverage (<code>measured</code>)
K6	Data	CF4 WF curl-suppression no-go (<code>measured_no_go</code>)

Sector-null precision. The two unreachable sectors are *not* the same kind of null. W^2 is a genuine, order-independent *structural* null: radial peculiar velocities satisfy $n^a \Omega_{ab} n^b \equiv 0$ exactly and CMB temperature is curl/Weyl-blind at EGS order; it re-opens only via a *different*

observable (transverse velocities, CMB B -modes). Ω_k is a leading-EGS-order *no-channel* that re-opens *beyond* leading order within the same channels.

3 Local boost vs global tilt: the kinematic deprojection (Axis C)

The vulnerability. A naive low- ℓ estimator that reads all quadrupole power as shear is not safe once the observer moves. With $\mu = \hat{\mathbf{n}} \cdot \hat{\mathbf{v}}$, the Doppler/aberration transform of the monopole T_0 carries $\mu^2 = \frac{2}{3}P_2(\mu) + \frac{1}{3}$, so a boost β deposits an $O(\beta^2)$ *kinematic quadrupole* aligned with the boost axis. A bulk-flow-only universe ($\Sigma^2 = 0$) would then read a false $\Sigma^2 > 0$, and the graded-comparator covariance is *not* diagonal: $F_{\Sigma^2, \Omega_{\text{tilt}}} \propto \beta^2 \neq 0$.

The fix. Because the tilt is linear in the velocity ($\Omega_{\text{tilt}} = \kappa_T \beta$), the boost quadrupole equals $\alpha(\Omega_{\text{tilt}})^2$ at the quadrupole-amplitude level, with α a β -independent constant of the registered response,

$$\alpha = \frac{4}{9} \frac{T_0^2 N_2}{\kappa_T^2 R_\Sigma}, \quad \tilde{\Sigma}^2 = \Sigma^2 - \alpha (\Omega_{\text{tilt}})^2, \quad (2)$$

so $\tilde{\Sigma}^2 \rightarrow 0$ on a pure-bulk-flow sky and $\tilde{\Sigma}^2 = \Sigma^2$ when $\Omega_{\text{tilt}} = 0$ (no over-subtraction). The physical origin is the ε_1^2 dipole-squared term of the boost kernel (coefficient +1 in the registered ε -normalisation). This is a *separate diagnostic surface* (http://obsstat/egs3_kinematic_deprojection.py): it never touches the bit-identical comparator x_C or the frozen registered low- ℓ statistics, and adds no new registered statistic. The content is gate-verified (EGS3-C1–C4) and the closed-form identities Wolfram-verified (wolfram/egs3_boost_tilt_separation.wls, 11/11):

- **Coupled Fisher (C2).** $F_{\Sigma^2, \Omega_{\text{tilt}}} \propto \beta^2$; the Σ^2 variance inflates by $1/(1 - r^2)$ (r the response correlation), returning to unity as $\beta \rightarrow 0$.
- **Identifiability (C3).** The boost and shear quadrupoles separate whenever their two $\ell = 2$ templates are independent: the Gram determinant $1 - P_2(\cos \angle)^2$ is positive unless the boost axis is aligned with the shear principal axis (then rank drops and $\tilde{\Sigma}^2$ over-subtracts).
- **Injection-recovery witness (C4).** A deterministic moment-level experiment (no map, no solver, no real data) shows the naive Σ^2 false-positive rate rises to ≈ 1 on a pure-boost sky while the deprojected $\tilde{\Sigma}^2$ stays at its nominal level (≈ 0.05), and genuine shear is recovered unbiased and covered.

This is a proven *estimator property*, not a detection: Σ^2 on the real sky stays **partial**, and $\tilde{\Sigma}^2$ supplies a boost-immune shear reading when data lands.

4 BASS-Extended joint PV+CMB analysis (Axis D)

The proposed joint likelihood $\ln \mathcal{L}_{\text{Total}} = \ln \mathcal{L}_{\text{PV}} + \ln \mathcal{L}_{\text{CMB}}$ pins Ω_{tilt} from peculiar velocities so the CMB inversion cannot leak power into a false Σ^2 . Its CMB *theory* sector is the native low- ℓ solver; the rest is ported honestly and runs now.

4.1 Feasible peculiar-velocity covariance (D1)

A dense $N \times N$ peculiar-velocity covariance is infeasible for the $\sim 38\text{k}$ CF4 groups ($N^2 \sim 1.4 \times 10^9$, $O(N^3)$ inversion). The coherent velocity field is dominated by a few large-scale modes, so

$$C_{\text{PV}} = \text{diag}(\sigma_v^2) + U \Lambda U^\top, \quad U \in \mathbb{R}^{N \times K}, \quad K \ll N \quad (3)$$

(the leading bulk+shear velocity modes) is inverted by Sherman–Morrison–Woodbury in $O(NK^2)$ (http://obsstat/pv_covariance.py; verified bit-for-bit against the dense solve). On the real CF4

catalogue this correlated-covariance bulk-flow GLS reproduces the K5 amplitude $|\mathbf{B}| \simeq 341 \pm 5 \text{ km s}^{-1}$ (measurement error; the cosmic-variance budget is separate). A full off-diagonal ξ_{ij} from a velocity power spectrum or a Wiener-filter field ensemble is a standing gate (`BLOCKED_MISSING_FIELD_REALI`). The leading bulk+shear model is the honest, feasible form.

4.2 JWST distance-anchor prior (D1, forecast)

A JWST-anchored distance prior — the CF4 groups cross-matched to published JWST distances (CCHP TRGB/JAGB, SH0ES Cepheids, TRGB-SBF) — shrinks the anchors’ distance errors and tightens Ω_{tilt} . A download pipeline (`dl_pipeline/scripts/download_jwst_anchors.py`, mirroring the CF4 acquisition pattern) fetches the authoritative IOPscience/arXiv tables; a committed cited seed keeps the analysis runnable offline, and `scripts/jwst_cf4_crossmatch.py` matches the anchors to CF4 (9 of 12 seed galaxies matched). Because these are nearby SN-host galaxies — which already have small peculiar-velocity errors — the current ~ 10 -anchor *global*-tilt gain is modest ($\sim 4\%$). This is reported honestly, as a labelled survey-design *forecast*, with the gain scaling shown against anchor fraction; it is not a measurement.

4.3 Coupled-Fisher degeneracy break (D2)

The tighter Ω_{tilt} precision enters the Axis-C coupled Fisher $f_{\Omega_{\text{tilt}}}$ diagonal and *strictly* reduces the Σ^2 covariance inflation $1/(1-r^2)$ — the solver-free, Wolfram-verified realization of “PV pins Ω_{tilt} so the boost cannot masquerade as shear” (`prior_precision_reduces_inflation`, PASS).

4.4 Real off-diagonal BipoSH measurement on SMICA (D3)

The off-diagonal $\langle a_{\ell m} a_{\ell' m'}^* \rangle$ signal *is* a real single-sky observable as the rotationally-invariant Bipolar Spherical Harmonic power $D_{\ell\ell'}^L$ (`htt/obsstat/biposh_smica.py`; exact Wigner-3j contraction). $L=1$ is the observer-boost/aberration $\ell \leftrightarrow \ell+1$ coupling; $L=2$ is the quadrupolar statistical-isotropy (SI) violation a global tilt or shear would source. Measured on the real Planck SMICA and Commander maps (`scripts/k1_biposh_smica.py`) and look-elsewhere-corrected against a matched isotropic GRF null (500 realizations), the global p -values ($\simeq 0.68$ SMICA / 0.65 Commander) are *consistent with isotropy*. This is the honest sense in which the off-diagonal $C_{\ell m, \ell' m'}$ is extracted from SMICA: the *data* off-diagonal is measured, while the *theory* $A_{\ell\ell'}^{LM}(g)$ prediction for a Bianchi g stays fail-closed. The FFP10/NPIPE E2E null (§7) is the pending upgrade.

4.5 The theory sector is fail-closed (D4)

The mode-coupled $C_{\ell m, \ell' m'}(g) / A_{\ell\ell'}^{LM}(g)$ prediction for a proposed Bianchi g requires the native low- ℓ solver (`AWAITING_NATIVE_LOWELL_SOLVER`). It is **fail-closed**: `joint_pv_cmb_forecast.anisotropic_cmb_covari` raises `OutOfScopeError` and never fabricates a covariance; the drop-in point is `bass.spectrum.cl.assembly.off_di`. Accordingly `evaluate_joint_loglike` returns a real PV log-likelihood and the real BipoSH data statistic but *withholds* any g -conditioned joint scalar, so the blocked sector is never silently zeroed. The solver-free substitute for the CMB side is the data-side BipoSH measurement (D3).

5 Real-data status

Real-data descriptors are computed on owned inputs; all are model-independent and carry no family/geometry/anisotropy-evidence claim.

- **K5 / D1 (CF4 bulk flow)**. A weighted-GLS estimator on the real Cosmicflows-4 group release (Tully et al. 2023, 38,049 groups) gives $|\mathbf{B}| \simeq 341 \pm 102 \text{ km s}^{-1}$, the error budget

cosmic-variance-dominated — high but not anomalous. The feasible correlated-covariance (Woodbury) tilt of §4 reproduces this amplitude. Coverage (0.67 vs 0.19 measurement-only) is geometry-and-error matched and *conditional* on a fixed Λ CDM $\sigma_{\text{cv}} = 150 \text{ km s}^{-1}$ per-component prior.

- **K6 (CF4 curl/vorticity).** The affine velocity-gradient decomposition of the real CF4++ Wiener-filter field returns a vorticity at most 0.55% of the shear at every radius, with an injected solid-body rotation recovered to machine precision. This is a WF mean-field curl-suppression *structural no-go*, not a vorticity measurement.
- **K1 (low- ℓ CMB morphology).** Stacking six registered low- ℓ statistics on the real Planck PR3 maps into a frozen max-scan gives a look-elsewhere-corrected global $p = 0.097$ (SMICA) / 0.121 (Commander) under an isotropic Λ CDM (GRF) null over 2000 realizations. The $\sim 25\%$ method dependence is itself a foreground/cleaning systematic; the two are reported side by side and *not* averaged. *This is the measurement the Planck-raw plan of §7 upgrades.*
- **D3 (low- ℓ CMB off-diagonal BipoSH).** A companion real-data channel to K1 that reads the *off-diagonal* $\ell\text{--}\ell'$ structure directly (§4); SMICA/Commander global $p \simeq 0.68/0.65$, consistent with isotropy.

Joint comparator (PR08-006). Assembling the discharges into g on real data gives a data rank of two: Ω_{tilt} is measured (K5 bulk flow) and Σ^2 is partial (K1 look-elsewhere only), while W^2 and Ω_k are fail-closed (K6 no-go; no leading-order channel) — never zeroed. No combined scalar x_C is formed, no MIO certificate is used as odds, and no scalar is promoted to a family. The rank-2 count is *one fully measured plus one partial* sector, not two measured sectors.

6 External-audit revision

Two independent external audits (a research-content review, verdict MINOR REVISIONS; and a claim-firewall re-review, verdict MAJOR REVISIONS) were folded in. The substantive items — the W^2 vs Ω_k null distinction, the e-value’s Λ CDM-null idealisation with an add-one finite-null $\hat{\alpha} = (k+1)/(n+1)$, the closure-conditional/registered-response scoping of the Fisher floor and shear bracket — were realised at the module level and propagated into all reporting surfaces. The genuine over-claims in the data labels were downgraded to the conditional/structural language used above. A permanent claim linter over the report, ledger, and blocker dossier forbids the flagged phrasings, and an independent re-check script reproduces the bracket constant, the e-value Markov bound, the Fisher-floor descent, the radial-vorticity identity, and the Ω_k genuine-zero demonstration. The subsequent “Leaky Universe” review — which raised the boost→shear leakage — is answered by Axis C (§3).

7 Planck raw-data analysis plan

7.1 Objective and the standing blocker

The K1 look-elsewhere correction is discharged on the real map under an idealised isotropic Λ CDM (GRF) null. What remains (BLOCKED.MISSING.PR4.E2E.ACCESS) is to replace that null with a **matched end-to-end (E2E) simulation ensemble** carrying real instrument noise, residual foregrounds, and the component-separation transfer, so that both the K1 morphology statistics *and* the D3 BipoSH statistics are reported with instrument-and-foreground systematics. The ensemble is served only through the Planck Legacy Archive (PLA) interactive query portal (and NERSC-authenticated paths), not as a plain-URL download; *cobaya* provides the Blackwell–

Rao C_ℓ -level low- ℓ likelihood, not the $a_{\ell m}$ map ensemble the morphology and BipoSH statistics require. The acquisition is therefore manual; everything downstream is implemented.

7.2 Data products and acquisition

The recommended route (matching the SMICA observed map) is the FFP10 component-separated simulation library:

- `dx12_v3_smica_cmb_mc_<00000..00999>_raw.fits` — nominally 1000 CMB (signal) MC, of which **999 are usable**: realization 00970 is a known missing/corrupt file on the PLA (ESA/PLA confirmation pending);
- `dx12_v3_smica_noise_mc_<00000..00299>_raw.fits` — 300 noise/systematics MC.

The E2E null is CMB+noise, paired *by parsed MC id*, `cmb[id] + noise[id mod  $n_{\text{noise}}$ ]`; id-based pairing is used so the missing 00970 leaves a single absent realization rather than misaligning every later pairing. The ensemble is therefore the **PLA-available** set (999 usable CMB MC + 300 noise MC), not all 1000. At full IQU $N_{\text{side}} = 2048$ this is ~ 1 TB; a dedicated 2 TB nvme accommodates the full set. Raw maps stay outside version control; only compact summaries and provenance are committed.

7.3 Pipeline and runner modes

The frozen pipeline is identical to the observed-map path: read each simulation (temperature only), downgrade-on-read to the analysis N_{side} , apply the matched low- ℓ mask, compute the six registered morphology statistics (and, for the D3 channel, the $L=1, 2$ BipoSH powers), and feed the observed vector and the simulation matrix to the look-elsewhere max-scan, which returns the +1-corrected global rank p . Processing is streaming (one map at a time), so disk and wall-time, not RAM, are the constraints. Three modes in `scripts/k1_global_maxscan.py` each write a *separate* artifact so the canonical GRF result is untouched (Route 4 noise-only; Route A full E2E, the exit gate; and a re-registered v2 precision set adding `proc- $N_{\text{side}}$ ,  $\ell_{\text{max}}$ , and a mask+inpaint`). The BipoSH channel (`scripts/k1_biposh_smica.py`) reuses the same GRF-now / FFP10-pending null path and the same frozen statistic set.

7.4 Higher-precision (v2) statistic set

On a 12-core/64 GB workstation the analysis is I/O-bound, so precision can be raised at essentially the same wall-time. The v2 set (`htt/obsstat/lowell_precision.py`) is a deliberately re-registered statistic set with three levers: processing N_{side} (default 64; the ceiling for $\ell \leq 8$, where $N_{\text{side}} = 16$ is Nyquist-sufficient and the only gain is removing the pixel-window suppression, Table 2); ℓ_{max} (default 30; the ℓ -summed statistics gain genuine information, the quadrupole–octupole alignments are intrinsically $\ell = 2, 3$); and a galactic mask + diffuse inpainting applied *identically* to the observed map and every simulation, so the mask response is absorbed into the null.

Table 2: Measured pixel-window w_ℓ (1.0 = no suppression). $N_{\text{side}} = 64$ removes the $N_{\text{side}} = 16$ suppression at $\ell = 6\text{--}8$; higher N_{side} adds nothing at $\ell \leq 8$.

N_{side}	w_2	w_6	w_8
16	0.9989	0.9924	0.9870
64	0.9999	0.9995	0.9992
128	1.0000	0.9999	0.9998

7.5 Compute budget

Parallelism (`--jobs`) fans the per-simulation loop over processes (`spawn` context, one thread per worker, avoiding the fork-after-OpenMP deadlock); the noise set is pre-downgraded once and cached. The measured per-map cost at v2 is ≈ 3 s; the PLA-available 999+300 ensemble is I/O-bound (a ~ 3.5 min nvme read floor), giving ~ 5 – 7 min wall-time on a Ryzen 5900X with `--jobs 12`. Peak memory is ≈ 1 GB per worker, so 12 workers use ≈ 12 GB — well within 64 GB. No additional RAM is required.

7.6 Exit gate and kill switches

K1 re-points `measured.partial`→`measured` only through the ledger generator, with a global rank p under the *matched* E2E ensemble; a configuration hash bound to {statistic list, mask, beam, ℓ -range, method}; an ensemble provenance manifest; and both component-separation methods reported side by side. The run stops if the map/mask/beam path differs between observed and simulations, if methods are mixed without labels, if fewer simulations are available than declared, if the statistic set is unfrozen, or if any text would claim E2E global significance without the matched provenance.

7.7 Co-evolution with the native low- ℓ solver

The v2 analysis uses $\ell = 2$ – 30 ; for internal consistency the future native low- ℓ atlas should predict the *same* statistics over the same range (Bianchi families, notably VII_h, diverge most at intermediate ℓ , and the shear Fisher floor keeps improving to $\ell \sim 20$). The recommended atlas ceiling is $\ell = 2$ – 30 (staged $\ell \leq 20$ then $\ell \leq 30$), conditioned on the exact-FLRW-first gate and per- ℓ convergence. This is separate (PR10) work and grants no detection authority by itself; the theory- g BipoSH prediction (§4, D4) is one of its consumers.

8 Claim envelope

The defensible envelope is a diagnostic methods-and-calibration framework: a rank-aware, provenance-bound sector comparator; a boost-immune shear estimator with a proven false-positive-rate and a coupled-Fisher error model; conditional EGS-type theorems under stated hypotheses; a feasible correlated-covariance bulk-flow analysis; and real, null-calibrated low- ℓ CMB descriptors (morphology and off-diagonal BipoSH). It does *not* establish a globally significant low- ℓ anomaly, an anisotropic geometry, a Bianchi-family identification, a native-solver validation, or a physical-vorticity detection; no diagnostic is used as posterior odds, no scalar is promoted to a family, the theory- g CMB likelihood is fail-closed, the JWST prior is a labelled forecast, and Σ^2 stays *partial*. The Planck-raw E2E analysis narrows the K1/D3 systematics gap; it does not change these boundaries.

A Reproducibility (commands)

Full Route-A E2E, v2 precision, parallel:

```
python scripts/k1_global_maxscan.py \  
  --cmb-mc-dir <mc> --noise-mc-dir <noise> --method smica \  
  --max-sims 1000 --precision --proc-nside 64 --lmax 30 --jobs 12
```

Pre-solver real-data + forecast surface (runs now):

```
SMICA BipoSH SI: python scripts/k1_biposh_smica.py  
CF4 Woodbury tilt + JWST forecast: python scripts/bass_extended_joint_forecast.py  
JWST acquisition: python dl_pipeline/scripts/download_jwst_anchors.py  
JWST→CF4 cross-match: python scripts/jwst_cf4_crossmatch.py
```

theorem gates: `make egs3-gates` (61 tests)
consolidated ledger: `python scripts/build-egs-results-table.py`

Other entry points:

download/run guide: `docs/research_program/K1_E2E_DOWNLOAD_GUIDE.md`
standing blockers: `docs/research_program/BLOCKERS.md`